Emeline Lavaux’s R Project

Section 1: Description of the relationship of interest

1) What are the two variables X and Y you have chosen to study? What is the X? What is the Y ?

I have decided to study the relationship between net greenhouse gas emissions and life expectancy by age and sex.

- X = Life expectancy by age and sex :

This is the average number of years a person is expected to live based on their age and sex. Life expectancy is a key indicator of public health and is influenced by factors such as healthcare access, environmental quality, and socioeconomic conditions.

- Y = Net greenhouse gas emissions :

This refers to the total emissions of greenhouse gases, such as carbon dioxide, methane.. These emissions contribute to climate change, which has wide-ranging effects on ecosystems, economies, and human health. This variable is measured intonnes per capita.

2) Why is it interesting to study the relationship between your X and your Y?

We often focus on how greenhouse gas emissions impact life expectancy, but in my project, I aim to flip this perspective by exploring whether valuable insights can be gained by examining how life expectancy influences greenhouse gas emissions.

Longer life expectancy typically leads to higher lifetime resource consumption, which increases greenhouse gas emissions due to greater demand for energy, transportation, housing, and healthcare. In countries with higher life expectancy, urbanization and industrial growth further drive emissions, as these processes are energy-intensive. Understanding how longevity affects emissions is essential for shaping policies focused on sustainable development, such as enhancing energy efficiency. Furthermore, rising life expectancy in developing nations could contribute to greater emissions, highlighting the need to address global inequalities in health and environmental impacts.

I think that studying how life expectancy affects greenhouse gas emissions is vital for understanding the environmental consequences of demographic shifts. As populations live longer, the relationship between consumption, energy use, and emissions changes, offering valuable insights into how societies can balance long-term well-being with environmental sustainability.

Section 2 : Description of your data

3) How is measured your variable X?

My variable X, life expectancy by age and sex, is measured in years. It represents the average number of years a person is expected to live from a specific age, assuming current mortality rates remain constant throughout their lifetime. The data is broken down by age group and gender and is calculated using mortality rates across various European countries.

4) How is measured your variable Y?

My variable Y, net greenhouse gas emissions, is measured in tonnes per capita. This metric represents the total greenhouse gas emissions produced by a country, divided by its population, giving an average emissions figure for each individual. It accounts for different types of greenhouse gases (CO_2, methane, etc.) by converting their global warming potential into equivalent amounts of CO_2.

5) What variable(s) did you use to combine your datasets on X and Y?

I combined the two datasets using the common variables Year and Country. Because the life expectancy dataset contained more countries than the greenhouse gas emissions dataset, I removed the countries that were not common to both datasets before merging. This allowed me to get a dataset that aligns both life expectancy and greenhouse gas emissions by country and year.

6) What is the unit of observation in your final dataset?

The unit of observation in my final dataset is at the Country-Year level. This means that each row represents a specific country's life expectancy and net greenhouse gas emissions for a particular year from 2014 and 2022.

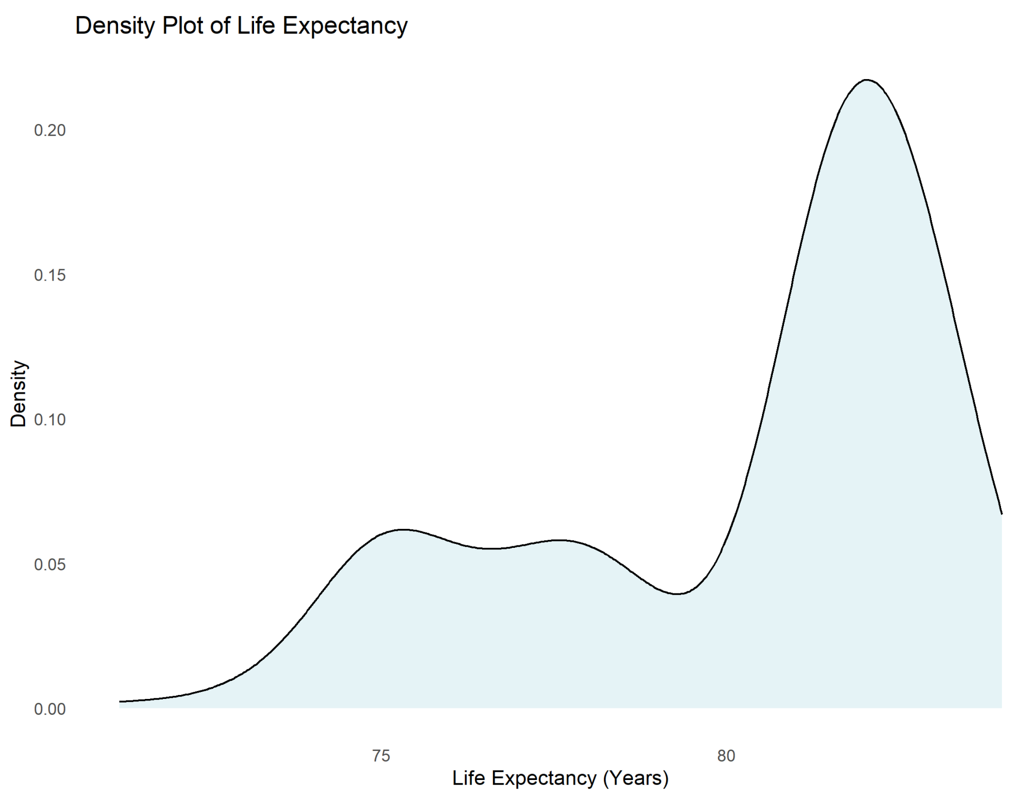
7) How many observations do you have in your final dataset?

In my final data set I have 270 observations.

Section 3 : Descriptive statistics

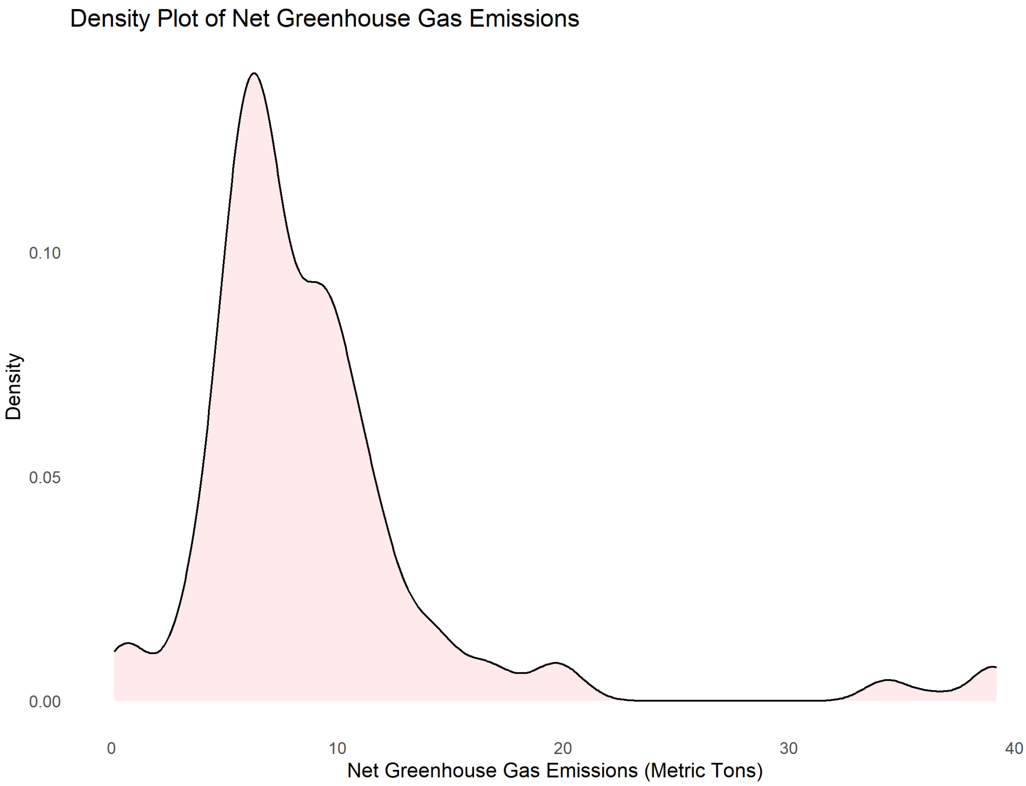
8) What is the distribution of your variable X? (Include an histogram if X is a categorical variable, include a density plot if X is a continuous variable). Make this graph as clean and readable as possible. Include this graph to this document.

My variable X is continuous and here is its density plot :



9) What is the distribution of your variable Y? (Include an histogram if Y is a categorical variable, include a density plot if Y is a continuous variable). Make this graph as clean and readable as possible. Include this graph to this document.

My variable Y is continuous and here is its density plot :



10) Draw a table of summary statistics with mean, standard deviation, minimum and maximum for both your variables X and Y. Make this table as clean and clear as possible. Include this table into this document.

Table 1: Summary Statistics of Key Variables

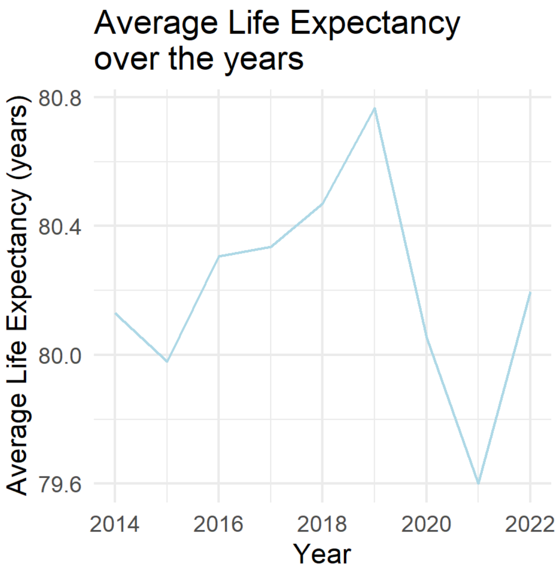
Variable	Mean	Standard Deviation	Minimum	Maximum
Share of greenhouse gas emissions	9.11	6.34	0.1	39.2
Life Expectancy	80.20	2.96	71.2	84.0

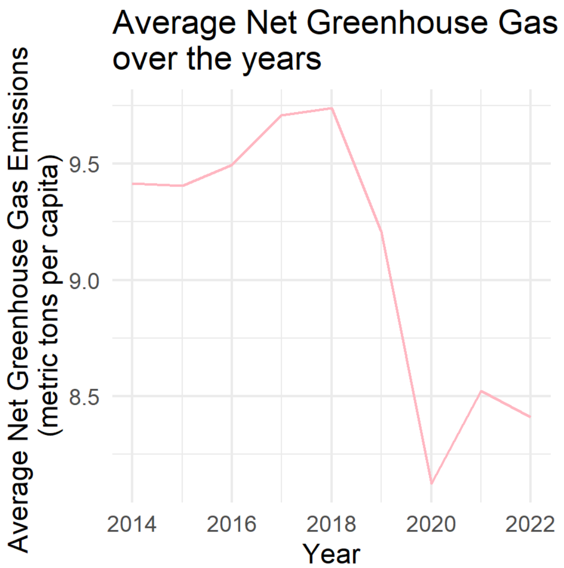
11) Draw a table of summary statistics where each row corresponds to 1 country and reports the name of the country, its mean for variable X over the entire period, its standard deviation for variable X over the entire period, its mean for variable Y over the entire period, and its standard deviation for variable Y over the entire period. Make this table as clean and clear as possible. Include this table into this document.

Table 1: Summary of Life Expectancy and Greenhouse Gas Emissions by Country

Country	Mean_X	SD_X	Mean_Y	SD_Y
Austria	81.57778	0.2635231	8.6111111	0.8268078
Belgium	81.54444	0.4034573	10.3222222	0.5717906
Bulgaria	74.18889	1.2474418	7.0333333	0.5338539
Croatia	77.81111	0.5688683	5.2555556	0.2743680
Cyprus	82.16667	0.5147815	10.5333333	0.5431390
Czechia	78.74444	0.6482883	11.8888889	0.3689324
Denmark	81.15556	0.3320810	8.7555556	0.9422196
Estonia	78.16667	0.6103278	14.0222222	3.2092748
Finland	81.67778	0.3073181	9.0000000	1.0210289
France	82.61111	0.2666667	6.2555556	0.4186619
Germany	80.98889	0.2147350	10.2888889	0.8922506
Greece	81.27778	0.5142416	8.1444444	0.9153020
Hungary	75.80000	0.7000000	5.9222222	0.2773886
Iceland	82.74444	0.4156655	37.1666667	2.3243279
Ireland	82.13333	0.4949747	13.9111111	0.7356025
Italy	83.02222	0.4236088	6.6555556	0.2920236
Latvia	74.77778	0.7479602	5.9666667	1.1937336
Lithuania	75.30000	0.7632169	4.7666667	0.4924429
Luxembourg	82.48889	0.2976762	18.4777778	2.1387951
Malta	82.32222	0.2333333	5.2777778	0.8870989
Netherlands	81.72222	0.2488864	11.3666667	1.1423660
Norway	82.73333	0.3741657	7.3555556	0.6366143
Poland	77.31111	0.8738484	9.5222222	0.3562926
Portugal	81.52222	0.2166667	6.3111111	1.2965767
Romania	74.81111	0.8447551	3.4666667	0.1658312
Slovakia	76.90000	0.9178780	6.3777778	0.5449261
Slovenia	81.13333	0.3391165	8.2111111	0.7007932
Spain	83.27778	0.4576510	5.9666667	0.5894913
Sweden	82.64444	0.3844188	0.6888889	0.4807402
Switzerland	83.57778	0.3562926	5.9222222	0.5093569

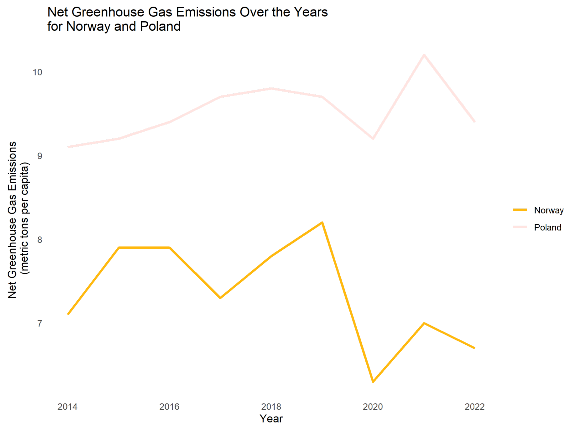
12) Plot the average evolution for variables X and Y over the years into two separated graphs. Make these graphs as clean and readable as possible. Include these graphs to this document.

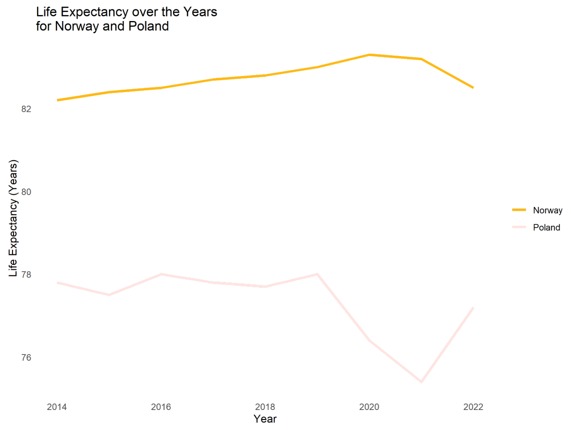




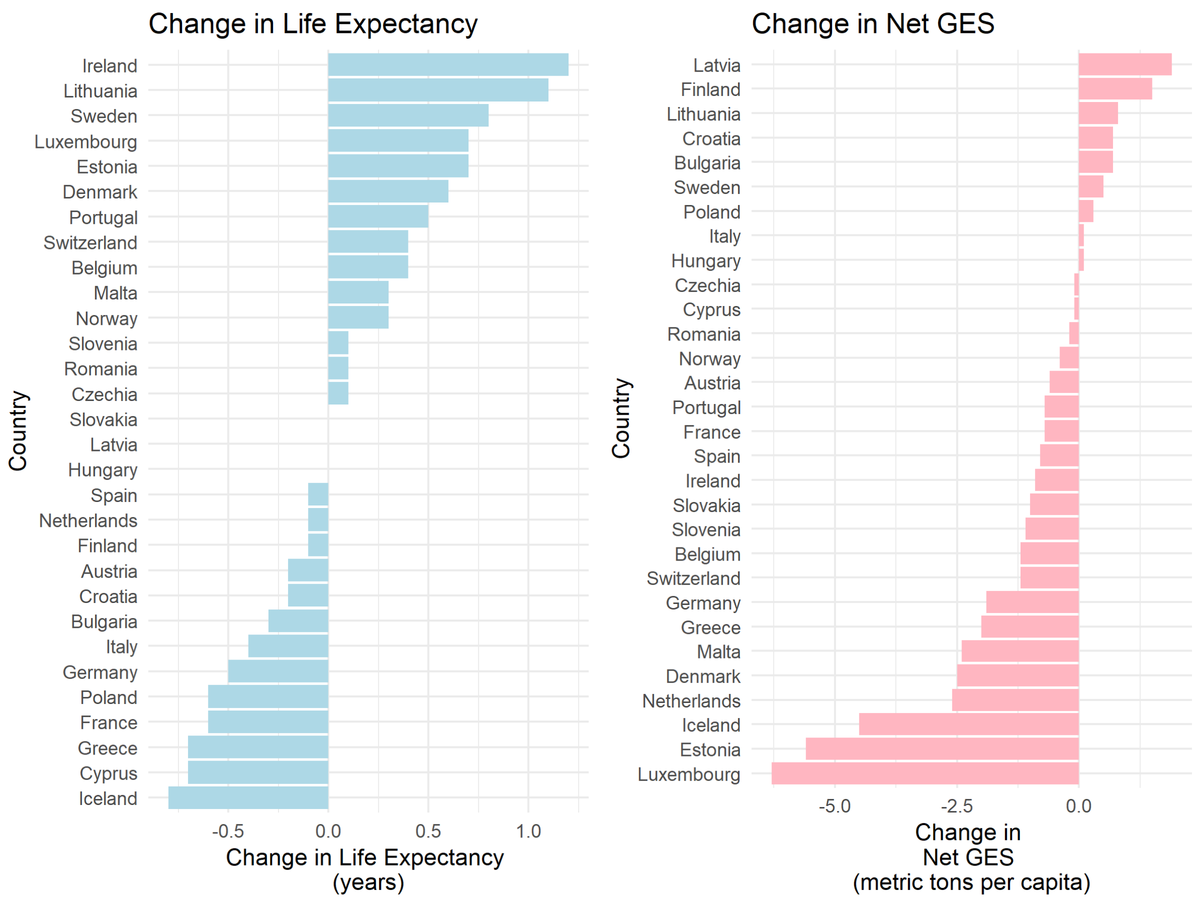
13) Select two countries and plot the average evolution for variables X and Y over the years for each one of these countries into two separated graphs. Make these graphs as clean and readable as possible. Include these graphs to this document. Explain why you selected these two countries and why they provide important insights regarding the relationship between X and Y.

Norway and Poland are interesting to compare because Norway has a high life expectancy and low greenhouse gas emissions per capita, largely due to its strong investment in renewable energy. On the other hand, Poland has lower life expectancy and higher emissions, driven by its reliance on coal for energy. This comparison highlights how different energy policies and economic approaches impact both health and environmental sustainability. It also shows the connection between fossil fuel dependence and lower life expectancy.



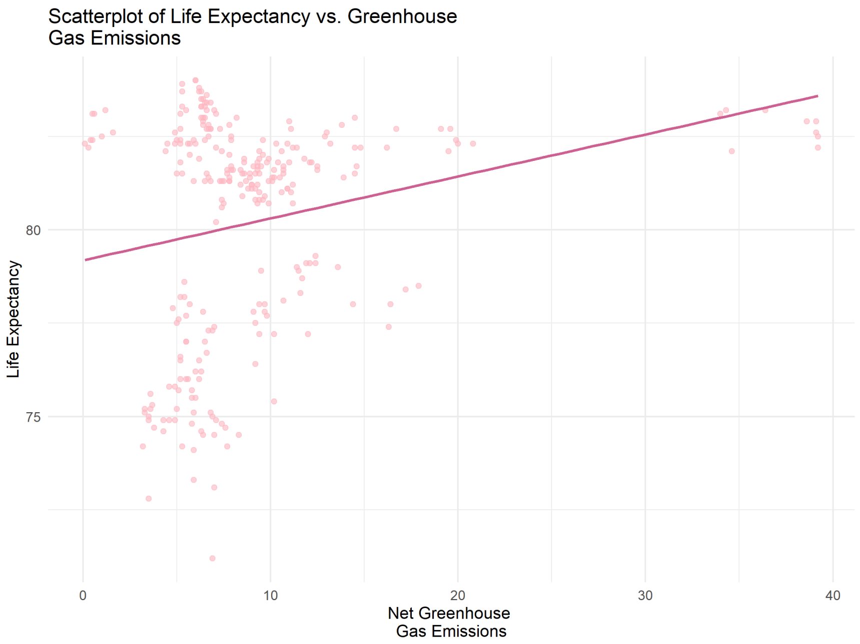


14) Compute change in X between first and last year for each country. Rank countries into a bar graph from the country with maximum value to the country with minimum value. Do the same with variable Y. Include the two graphs into 2 separated panels of a same graph. Make this graph as clean and readable as possible. Include this graph to this document.

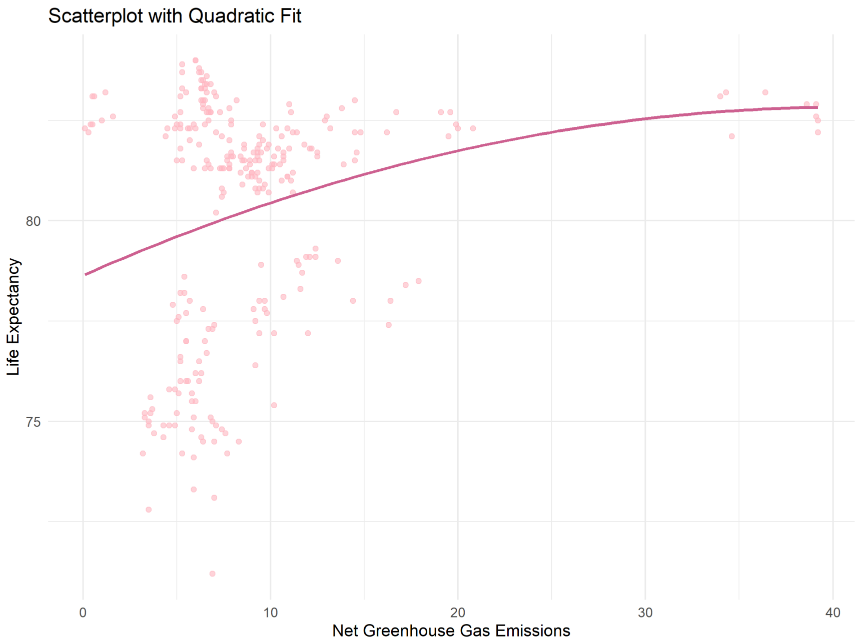


Section 4: Relationship between X and Y

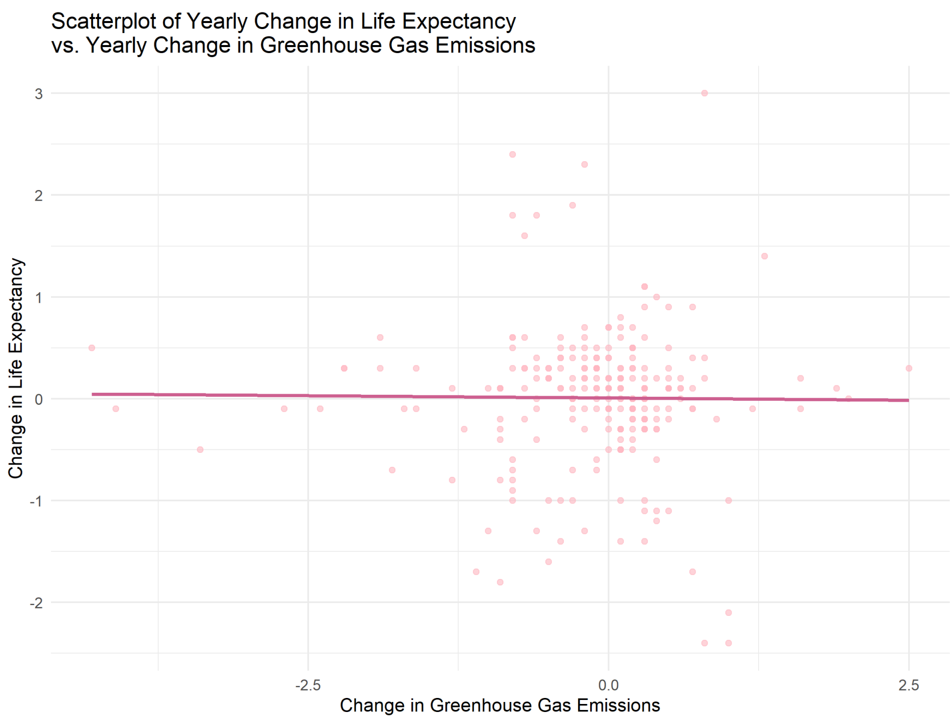
15) Plot a scatterplot for the relationship between Y an X. Include a linear fit of the data on the graph. Make this graph as clean and readable as possible. Include this graph to this document.



16) Plot another scatterplot of the relationship between Y and X. Include a quadratic fit of the data on the graph. Make this graph as clean and readable as possible. Include this graph to this document.



17) Compute within-country changes over the year for variables X and Y. Plot a scatterplot between the within-country change in Y and the within-country change in X. Include a linear fir of the data on the graph. Make this graph as clean and readable as possible. Include this graph to this document.



18) Regress your variable Y on your variable X. Export the results into a table and include the table to this document. Make this table as clean and clear as possible

19) Regress your variable Y on your variable X controlling for country fixed effects. Export the results of question 18 and 19 into a same table and include this table to this document. Make this table as clean and clear as possible.

20) Regress your variable Y on your variable X controlling for year and country fixed effects. Export the results of question 18, 19 and 20 into a same table and include this table to this document. Make this table as clean and clear as possible.

Table 1: Regression of Net Greenhouse Gas Emissions on Life Expectancy

	Dependent variable:		
	Net Greenhouse Gas Emissions		
	Model 1	Model 2	Model 3
	(1)	(2)	(3)
Life Expectancy	0.515***	−0.007	−0.480***
	(0.127)	(0.124)	(0.125)
factor(Country)Belgium		1.711***	1.695***
		(0.504)	(0.410)
factor(Country)Bulgaria		−1.631	−5.125***
		(1.046)	(1.009)
factor(Country)Croatia		−3.382***	−5.164***
		(0.687)	(0.624)
factor(Country)Cyprus		1.926***	2.205***
		(0.509)	(0.417)
factor(Country)Czechia		3.258***	1.917***
		(0.614)	(0.541)
factor(Country)Denmark		0.141	−0.058
		(0.507)	(0.413)
factor(Country)Estonia		5.387***	3.773***
		(0.658)	(0.591)
factor(Country)Finland		0.390	0.437
		(0.504)	(0.410)
factor(Country)France		−2.348***	−1.859***
		(0.520)	(0.430)
factor(Country)Germany		1.674**	1.395***
		(0.509)	(0.417)
factor(Country)Greece		−0.469	−0.611
		(0.505)	(0.412)
factor(Country)Hungary		−2.730**	−5.463***
		(0.876)	(0.829)
factor(Country)Iceland		28.564***	29.116***
		(0.524)	(0.435)
factor(Country)Ireland		5.304***	5.567***
		(0.509)	(0.416)
factor(Country)Italy		−1.945***	−1.262**
		(0.535)	(0.448)
factor(Country)Latvia		−2.693**	−5.909***
		(0.982)	(0.942)
factor(Country)Lithuania		−3.889***	−6.859***

1

Our model reveals a complex relationship between life expectancy and greenhouse gas emissions. Initially, it appears that longer life expectancy is linked to increased emissions (Model 1), but after accounting for country-specific factors (Model 3), this relationship reverses, suggesting that higher life expectancy might actually be associated with lower emissions. This finding challenges the assumption that longer life spans necessarily lead to greater environmental degradation.

Future research could enhance the model by incorporating additional demographic and economic factors such as GDP per capita, education levels, and urbanization rates, to better understand how these variables interact with life expectancy to influence emissions. These insights could be valuable for designing policies that promote both long-term human well-being and environmental sustainability.